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Math 3108 – Linear Algebra
Section 101 – Brigham
Exam 2
March 13, 2023

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing “BPP” on the relevant problem.
- 10) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	4	5	Sum
Points Earned						
Maximum Points	15	20	21	22	22	100

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

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Multiple Choice and True/False. Circle the correct answer.

1. (10 pts) Let A and C be a 5×5 matrices. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Label the following as True or False.

i) _____ The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

ii) _____ $\det A = 0$.

iii) _____ The column space of A is isomorphic to $\mathbb{R}^5$.

iv) _____ A has 5 pivot positions.

v) _____ $CA = I$

2. (2 points) The set $\{a + bt : a, b \in \mathbb{R}\}$ is a subspace of $\mathbf{P}_3$.

i) Always true

ii) Sometimes true

iii) Never true

3. (3 points) Let A be an $n \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^n$?

i) $\text{Col } A$ (the column space of A)

ii) $\text{Nul } A$ (the null space of A)

iii) Both $\text{Col } A$ and $\text{Nul } A$.

iv) Neither $\text{Col } A$ nor $\text{Nul } A$.

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Open Answer. Show ALL your work!

4. (8 points) Use the method of your choice to calculate A^{-1} given $A = \begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix}$. Then, use the inverse you found to solve $A\mathbf{x} = \mathbf{b}$ given $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$.

5. (12 points) Find the LU -factorization of the matrix $A = \begin{bmatrix} 1 & -2 & -4 & -3 \\ 2 & -7 & -7 & -6 \\ -1 & 2 & 6 & 4 \end{bmatrix}$.

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6. (12 points) Compute the adjugate of the matrix $A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 4 \end{bmatrix}$.

7. (9 points) Use the LU factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ to solve $A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$

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8. Consider the matrix $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$.

a) (9 points) Use any valid combination of cofactor expansion and/or row operations to calculate the determinant $\det A$.

b) (4 points) Find $\det A^T$ using as few steps as possible.

9. (9 points) Let $W = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} : a, b, c \in \mathbb{R}, \text{ and } a + b + c = 0 \right\}$. Is W a subspace of $\mathbb{R}^3$?

Explain.

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10. The matrices A and B given below are row equivalent. (You do not need to check this.)

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a) (5 points) Find a basis for the column space of A .

- b) (10 points) Find a basis for the null space of A .

11. (7 points) Determine whether the set of vectors $B = \{(1-t)^2, t-t^2, 1-3t+2t^2\}$ forms a basis for $\mathbf{P}_2$.